

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1195

Mountaineer Wind Energy Center 138 kV
March 18, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1195. The Interconnection Customer proposes a 238.5 MW wind facility interconnecting at the Mountaineer Wind Energy Center 138 kV substation in the West Virginia area. The request was received on April 27, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 319 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$28,748,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ronkonkoma - Comerio 9703 138 kV	Greenland Gap - Santee Cooper 345 kV	118.7	259	10.6
Shady Grove - Ronkonkoma 138 kV line	Greenland Gap - Santee Cooper 345 kV	116.5	275	10.6
Gold Beach - Green Bluff 138 kV	Greenland Gap - Santee Cooper 345 kV	114.4	332	15.3
Shady Grove Substation - Green Bluff 138 kV	Greenland Gap Substation - Santee Cooper 345 kV	113.1	296	10.6
Fitzwilliam - Gold Beach 138 kV	Greenland Gap - Santee Cooper 345 kV	107.3	213	10.0
Comanche - Fish Creek 138 138 kV	Greenland Gap - Santee Cooper 345 kV	103.7	400	5.9
Apple Inc - Mountaineer Wind Energy Ctr Substation 138 kV	Mountaineer Wind Energy Ctr - Santee Cooper 345 kV	102.6	255	36.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ronkonkoma - Comerio 9703 138 kV	Rebuild line with higher-capacity conductor	\$4,090,000	74% (\$3,048,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ronkonkoma Substation - Shady Grove 138 kV L	Rebuild line with higher-capacity conductor	\$11,615,000	74% (\$8,655,000)
Goldbeach - Green Bluff 138 kV	Rebuild line with higher-capacity conductor	\$3,562,000	79% (\$2,823,000)
Green Bluff - Shady Grove 138 kV	Rebuild line with higher-capacity conductor	\$2,261,000	74% (\$1,685,000)
Fitzwilliam - Gold Beach 138 kV	Reconductor line	\$2,785,000	74% (\$2,047,000)
Comanche - Fish Crk 138 138 kV	Reconductor line	\$4,123,000	40% (\$1,626,000)
Apple Inc - Mountaineer Wind Energy Ctr Substation 138 kV	Reconductor line	\$9,753,000	91% (\$8,864,000)
	Total allocated to project		\$28,748,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.